1. INTRODUCTION

1.1. INTRODUCTION TO ORTHOTROPIC STEEL DECK BRIDGES

Many of the world's most magnificent modern bridge structures utilize the orthotropic steel plate systems as one of the basic structural building blocks for distribution of traffic loads in decks and for the stiffening of slender plate elements in compression. Examples include the new San Francisco Oakland Bay Bridge, Self Anchored Suspension Span in California and the proposed Strait of Messina Bridge in Italy. Stiffened steel plates have been used for many years in a wide range of steel construction applications. They are particularly prevalent in the ship building industry and for hydraulic applications such as tanks, gates, and locks. The first orthotropic steel deck (OSD) bridge was developed by German engineers in the 1930's and the first such deck was constructed in 1936. In the United States, a similar system was built and often referred to as a "battle deck" because it was considered to be as strong as a battleship.

Generally, the OSD system consists of a flat, thin steel plate, stiffened by a series of closely spaced longitudinal ribs with support by orthogonal transverse floorbeams (Figure 1-1). The deck has considerably different stiffness characteristics in the longitudinal and transverse direction. Hence, the deck is considered to be structurally anisotropic. The name "orthotropic" arises out of a shortened form of the technical structural description of the system. According to *Orthotropic Bridges – Theory and Design* (Troitsky, 1987), "Because [the] ribs and floorbeams are orthogonal and because in both directions their elastic properties are different or anisotropic, the whole system became known as orthogonal-anisotropic, or, briefly, orthotropic." German engineers are credited with creating the word "orthotropic" and a patent was registered in 1948 (Sadlacek, 1987).

The OSD is efficient in that it is integral with the supporting bridge superstructure framing as a top flange common to both the transverse floorbeams (FBs) and longitudinal girders. This results in increased rigidity and material savings in the design of these components. As with other conventional steel-framed construction, loads are generally transferred by FBs transversely to the main load carrying system, such as longitudinal girders. Although, transverse members in the deck are most often referred to as *floorbeams*, there are also other commonly used terms such as crossbeam and/or diaphragm. Diaphragm is most appropriate when the deck is used in a redecking application and the member is rigidly, continuously attached to the existing bridge framing. Intermediate floorbeam may also be used where the FB member is not supported at its ends by main members and it is only serving to provide load distribution in the deck. *The defining characteristic of the OSD bridge is that it results in a nearly all steel superstructure* which has the potential (with minimal maintenance) to provide extended service life and standardized modular design, as compared to more conventional bridge construction.

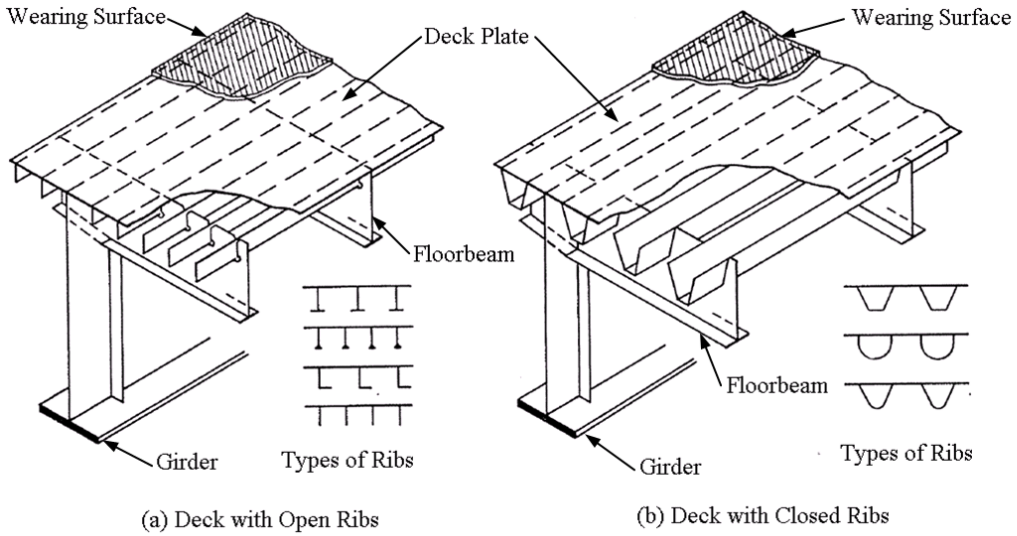


Figure 1-1 Components of the Orthotropic Steel Deck Bridge Girder System showing (a) Open Ribs and (b) Closed Ribs (AISC, 1963. Copyright American Institute of Steel Construction Reprinted with permission. All rights reserved.)

The OSD system has been utilized successfully for thousands of bridges worldwide, particularly in Europe, Asia, and South America. The United States has not yet fully embraced this technology, with only an estimated 100 such bridges in its inventory. The OSD bridge has been most commonly used in the United States for particular design conditions. One condition is for long span structures where the minimization of dead load is paramount, such as the new Tacoma Narrows Bridge. A second example is for box girders which contain slender compressive plate elements requiring stiffening, such as the Alfred Zampa Memorial Bridge (see Figure 1-2) A third example is for redecking of existing major bridges on urban arterials where rapid construction is needed, such as the Bronx-Whitestone Bridge.

OSD construction also has tremendous potential for use in short to medium span "workhorse" girder bridges when located on a high-volume roadway where accelerated construction or extended service life is required. There is a recent trend in the United States towards using bridge systems that are more rapidly constructible to minimize impacts to the traveling public, and to solutions that offer more long term durability and economy with the goal of 100 years of service life (Mistry and Mangus, 2006). The OSD bridge can provide an economical solution to meet these criteria. Furthermore, OSD is able to be constructed quickly because most of the components are prefabricated. Additionally, complete future redecking is rendered unnecessary, which minimizes any major traffic impacts in the future. Furthermore, the OSD provides a smooth continuous riding surface durable against deicing salts with minimal joints to prevent leakage and protect the other bridge components.



Figure 1-2 Erection of Orthotropic Box Girder Segment for the Alfred Zampa Memorial Bridge lifting a Deck Segment into Place

It is recognized that OSD bridges have not been problem-free historically. They present unique challenges in terms of design and construction as compared to conventional bridge construction. Fatigue cracking has been observed more frequently in such decks resulting from the complicated welded details combined with stresses that can be more difficult to quantify and, in particular, early designs which attempted to overly minimize plate thicknesses to reduce weight. In addition, the designs of critical details are not controlled by dead load or ultimate strength, but rather, live load (from individual wheels, in some cases). In any element where cyclic live load stress ranges dominate the design, fatigue will be the controlling limit state.

Early analytical tools were limited in their ability to quantify the stress states at these details and the early experimental fatigue resistance database was limited. Moreover, the fatigue performance of many of these details can be sensitive to fabrication techniques. Design and detailing practices relied heavily on experience gained through trial and error. Unfortunately, many trials were unsuccessful, creating questions among owners as to their long-term effectiveness in United States highway infrastructure. It is unfortunate that many of the reports of cracking have occurred in redecking projects where the interactions between new OSD and existing structure are difficult to account for, and design optimization is not easily achievable if at all possible.

The potential for cracking at the rib-to-deck plate (RD) weld is indicative of this problem. Whereas this one-sided weld was once a source of performance issues, it is now executed with a vast increase in consistency and performance by using a partial joint penetration (PJP) with controlled penetration, and with no melt thru allowed. Cracking is also possible at the rib-to-floorbeam (RF) intersections, where 3-dimensional stresses are generated by the in-plane flexure of the FB response combined with the out-of-plane twisting from the rib rotations. *All of these details have been the subject of extensive research efforts over recent decades, providing better understanding of performance and proper design*. For example, a stress-relieving cut-out in the FB around the rib (Figure 1-3) has performed well when the geometry is appropriately designed.

One clear advantage to the OSD is that it is a highly redundant system and minor cracking is often more of a nuisance to be observed and documented rather than a serious threat to the strength or integrity of the structure.



Figure 1-3 Orthotropic Deck Rib-to-Floorbeam (RF) Connection showing the Stress Relieving Cut-out Detail

Wearing surfaces applied to OSDs have also exhibited performance problems in the past with cracking, rutting, shoving, and/or delamination, which has often resulted in early maintenance and resurfacing. These problems have generally been attributed to inadequate construction control, environmental related degradation of the materials, or flexible design of the steel decking. This is also a symptom of the previous lack of experience. Recent research and development and general design improvements, such as minimum deck plate thickness of 14 mm to 16 mm (9/16 inch to 5/8 inch) have addressed the causes of many of these previous failures. Additionally, *current design concepts have proven successful in many modern OSD bridges in the United States and abroad*.

There are two broad categories of surfacing materials currently being used: (a) bituminous surfacing systems including mastic asphalts, latexmodified asphalts, and reinforced asphalt systems; (b) polymer surfacing systems, including epoxy resins, methacrylates and polyurethanes. Although not mandatory, many bituminous surfacing materials used on steel orthotropic deck bridges are 50 mm (2 inches) or greater in thickness, while most polymer surfacing materials are 20 mm (3/4 inch) or less in thickness. The climate generally dictates which type of surface is to be selected since bituminous surfaces are more sensitive to changes in temperature. Both wearing surface types have demonstrated a service life in excess of 30 years. As with all manner of bridges, no matter what type of wearing surface is utilized, regular maintenance and occasional resurfacing will be required during the design life of the deck.

The corrosion resistance of OSDs has historically been very good. The top side is protected by the wearing surface, and the bottom side can be protected with a conventional paint system. Similar to other steel bridge structures, it may require regular maintenance in terms of repainting. Moreover, OSDs are typically made continuous, without joints, for extended lengths, which

minimizes potential locations for water penetration. Thus, the coating on the underside of the deck can last for a long period unless it is subjected to direct saltwater spray. The individual ribs are typically sealed with end plates that prevent moisture from entering the interior of the rib. Outside the United States, a common approach has been to use a fully closed box girder cross-section and employ an in-service dehumidification system on the interior to essentially eliminate the possibility of corrosion, and thus there is no need for an interior paint system.

One of the primary reasons for advancement of OSD bridges in the United States is the application of modern techniques for engineering analysis and design. Orthotropic behavior is advanced since the plate distributes lateral loads in two directions and involves integral behavior of the deck with the FBs and girders. Engineers in the United States are generally unfamiliar with these types of structures and existing published manuals, such as (AISC, 1963) and (Troitsky, 1987), although state-of-the-art at the time of their copyright, have become dated. Additionally, early simplified analysis methods do not provide for a complete engineered design. *This Manual aims to gather and disseminate the modern OSD technology based on worldwide practice*. Emphasis is now placed on the fact that the design of these structures is generally controlled by fatigue limit states. Details necessary to make these structures work require advanced fatigue evaluation techniques that must rest on accurate stress range calculations, which is possible with the use of the Finite Element Analysis (FEA) and/or prototype testing. The fatigue testing database has grown considerably over the last few decades, which has provided the necessary data for proper evaluation and detailing for fatigue resistance.

Of equal importance to the engineering design are the means, methods, and quality control of the fabrication and construction. History has demonstrated that refined analysis and design can be rendered meaningless when the construction is not executed properly. Orthotropic details are also advanced in terms of fabrication and must be treated with care. To be successful, they require detailed construction specifications and quality control measures in production. As fabricators in the United States gain more experience in these projects, they will be viewed with less risk, which will promote more bid participation and lower cost. One factor that has exacerbated the risk for fabricators is the unwillingness of engineers to modify construction specifications to reduce risk and increase economy. This has generally been due to lack of data. This Manual provides the necessary information to solve this problem.

General consensus is that for OSD bridges to become cost effective in the United States, standardization is critical. *Standard panel details will promote repetition, economy in design, and fabrication as well as improve quality of the finished product*. Standards would also limit the need to conduct refined engineering analysis or prototype testing for every new project. However, optimum standard designs cannot yet be definitively established until more laboratory testing is conducted and domestically produced systems prove successful in service. This Manual promotes sound detailing concepts based on the current available knowledge, yet leaves room for advancements as more data becomes available and engineers discover improvements to existing designs. It is conceivable that after a period of development (which may last one or two decades) the application of OSD bridge decks will become second nature and standard designs will be available without the need to perform refined analytical procedures or laboratory testing. This Manual is seen as an initial step to achieve this ultimate goal.

OSDs can compete in cost with other suitable deck alternatives, such as the conventional cast-in-place concrete deck and concrete filled grids, when life cycle cost is considered. Because of their higher initial costs, OSDs must necessarily have lower maintenance needs and longer lives in order to be justified. Laboratory studies have demonstrated that an OSD with a 100-year design life may be achievable. Additionally, existing bridges like the Golden Gate Bridge and Fremont Bridge have shown that they are capable of performing with minimal required maintenance. With proper design and detailing, well executed fabrication, and quality construction, the full potential of OSD bridges can be exploited throughout the United States.

The target audience for this Manual includes engineers, owners, contractors, fabricators, and researchers in the bridge industry. The Manual gathers and summarizes worldwide knowledge from countries such as Germany, Denmark, France, Russia, Brazil, China, and Japan, and put it into context with United States design and construction practice. The project types covered in this Manual are long-span bridges, such as cable-stayed and box girder systems, bridge redecking, and decks for the more common "workhorse" girder bridge. It is the last case for which OSDs have not been used frequently in the United States, but with the potential for large volume and standardization of design, the Manual attempts to promote consideration in such cases.

1.2. ORGANIZATION OF THE MANUAL

This Manual covers the relevant issues related to OSD bridge engineering, including analysis, design, detailing, fabrication, testing, inspection, evaluation, and repair. Chapter 2 begins with discussion of some the various applications of OSD bridge construction to provide background with case study examples. Chapter 3 provides basic criteria for the establishment of a cost-effective and serviceable OSD bridge cross-section with detailing geometry that has been used on recent projects worldwide. Chapter 4 provides the relevant information necessary for the engineering analysis of the OSD bridge, including fundamental behavior and application of refined techniques. Chapter 5 outlines the requirements for complete design of OSD bridge superstructures by evaluation of applicable limit states using the Load and Resistance Factor Design (LRFD) methodology. Chapter 6 addresses design details such as materials, corrosion protection, minimum proportions, and connection geometry. Chapter 7 provides basic fabrication, welding, and erection procedures for OSD bridge components, illustrated by photos of shop and field practices. Chapter 8 provides recommended methods for maintaining and evaluating OSD bridges, including inspection, load rating by LRFR, rehabilitation strategies, and fatigue retrofit. Chapter 9 addresses all issues related to wearing surfaces. Chapter 10 covers testing of OSDs. The culmination of all the information provided is demonstrated in the design examples of Chapter 11 which contains one multi-girder bridge example and one cable-stayed bridge example demonstrating engineering design of OSD bridges by refined analytical techniques.

2. BRIDGE APPLICATIONS

2.1. GENERAL BACKGROUND

There appears to be a general lack of experience with orthotropic steel deck (OSD) bridges in the United States with the exception of a few major projects. As such, some time should be devoted to study of previous applications. Success in bridge design has proven to rely on experience and applying lessons learned from past projects. The historical lessons have allowed engineers and researchers to continually make improvements to OSD systems for decades, and since much of the worldwide experience with OSDs lies outside the United States, it is equally important to look internationally to gather the fullest range of examples.

This chapter is intended to highlight a few of the notable successful projects worldwide and let the readers follow through to discover more details. Projects from the Americas, Europe, Middle East, and Asia are discussed herein. Good summaries of OSD bridge projects can be found in the references by Troitsky (1985), Huang et al (2008), Hoorpah (2004), Korniyiv (2004), and Choi et. al. (2008). Although each governing body around the world has adopted subtle differences in terms of design and detailing practice, there is relative consistency as the collective knowledge has been shared in an unprecedented way.

The greatest advantage of OSD construction is realized when it is used in long-span bridges because it is comparatively lightweight, and it can work in composite action with the main longitudinal members. OSDs are also excellent candidates for bridges in seismic zones where they can reduce seismic inertia forces on piers and foundations and are able to undergo ductile deformations without sudden failures. OSDs are also highly desirable in movable bridges where they provide enduring performance with greater ease of movement. Additionally, their ability to perform structurally in a raised position better than other decks renders them superior. ODSs have been used in railway bridges, where the least maintenance, most durable solution is desired. While they are not effective at resisting direct train loads (especially where heavy locomotives are involved) they can perform well when ballasted for load distribution. In fact, many of the fatigue concerns discussed throughout this manual can be reduced or eliminated when ballast is utilized. In summary, OSD design can provide cost-effective solution in the following conditions:

- Long-span bridges
- Movable bridges
- Bridges in seismic zones
- Bridges where rapid construction is required
- Bridges where extended service life is required
- Cold weather conditions where cast-in-place concrete is difficult

Since many of the existing applications of orthotropic construction are long-span, signature bridges, they are all very special designs in terms of the bridge cross-section and superstructure configuration. The designs are often controlled by particular design criteria, constructability, or other considerations that in retrospect may not be obvious to the outside investigator. Prudence must be exercised when attempting to reuse any design detail that was used in a past project. For

instance, internal rib bulkheads have been used on recent bridge redecking projects to control certain secondary stresses due to unique interactions with the existing superstructure. Thus, it should not be assumed to be the ideal solution for new construction (see Chapter 4 for more on this subject).

In the following sections, a number of notable orthotropic bridge projects are discussed based on the general categories: Plate Girder Bridges, Box Girder Bridges, Suspended Span Bridges, Bridge Redecking, and Movable Bridges. Although there are many different bridge types that utilize orthotropic construction, it is emphasized that basic design of the OSD is quite similar in all cases. The commentaries provided are not intended to make judgments on previous designs based on present knowledge. They are instead intended to be discussions of typical examples of bridge types, and what they achieved at the time of their construction.

2.2. PLATE GIRDER BRIDGES

OSD bridges of this type were originally built in the 1930s under the name "battledeck floor" structures. A deck plate was welded to the tops of longitudinal I, T, or WF sections. The sections acted as ribs and were spaced up to 0.838 m (33 inches) on center, depending on the deck plate thickness. Transverse delivery of floor loads to the main girders was provided by floorbeams. Local load distribution, if deemed necessary in addition to that of the deck plate itself, was often provided by additional transverse plate stiffening welded to ribs and deck plate. Remnants of this type of construction still exist in prototypes such as the North Saginaw Road Bridge in Michigan (1920). Similar examples were produced in Germany with bent plate flanges over the Autobahn.

OSDs were used in the early 1950s for longer span plate girder bridges. Ribs always run longitudinally with transverse framing acting to distribute loads to several longitudinal girders and as support for the deck plate. Examples of these are the Kurpfalz Bridge in Mannheim and the Eddersheim Bridge Germany of 1950 and 1953, respectively.

The year 1956 saw a major achievement in the construction of this bridge type with a three-span continuous structure in Belgrade (the Save River Bridge). Two variable depth girders spanning 75 - 261- 75 m (246 - 856 - 246 ft) over two interior piers, supported a deck with flat plate ribs. The 25 mm by 267 mm (1 inch by 10.5 inch) ribs spanned continuously between FBs spaced at 1.585 m (5.2 ft) centers. The roadway width of this bridge was 11.98 m (39.3 ft).

The Golden Horn Bridge, erected in 1974 in Istanbul, consists of eight continuous spans of two plate girders, for a total of 819.3 m (2688 ft) and with the major span of 132.9 m (436 ft). The 24.7 m (81 ft) wide roadway deck is supported by an OSD consisting of closed ribs, 0.222 m (83/4 inches) deep, spaced at 0.622 m (24.5 inches) on center and spanning 4.50 m (14.75 ft) to FBs. Two plate girders, spaced at 24 m (78.75 ft), support the entire structure. Pedestrian walkways cantilever outboard of the plate girders.

In the United States, the use of OSDs did not take hold as it did in Europe. Instead, urban congestion demanded better use of space with smoother roadway alignment at bridge approaches and crossings. Curved bridges, which seemed to better meet this need and were built with the use of composite concrete decks on steel plate girders, would require complex details such as rib kinks to make OSD work. Three examples of orthotropic plate girder bridges used in the United

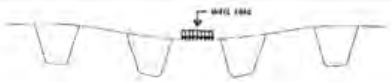
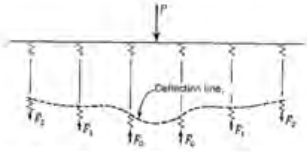
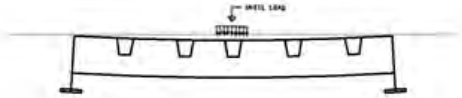
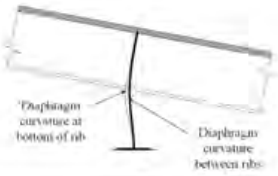
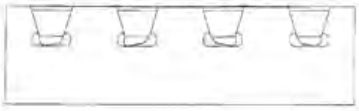
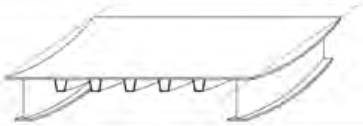
Therefore, it is recommended that the bulkhead be avoided when possible. As such, new designs should start without it. In cases of introducing a new OSD on an existing structure that offers little head room and requires an exterior cut-out, and has a relatively flexible FB, the bulkhead may help reduce stresses at the RDF sufficiently to make an OSD design viable.

4.2. BEHAVIORAL MECHANISMS

The conventional method of bridge analysis is to assume that the complete structural system is comprised of several basically independent sub-elements such as the deck, stringers, FBs, and primary superstructure. This approach is based on the assumption that each element acts relatively independently and transfers load from itself to the next element without consideration of the real interaction among the various sub-elements. For most applications, this has shown to yield conservative designs and can be implemented with a simplified design methodology. In the OSD bridge, the deck plating, ribs, FBs, and main girders are all integrated into one structural unit. The deck panel must perform several functions simultaneously, including distribution of wheel loads and acting as the top flange of both the FBs and main girders. The fact that the OSD performs all these functions results in a very efficient utilization of material but the interactions cannot be ignored.

In the mid 1900s, prior to the widespread availability of modern FEA software, it was proposed to decompose the OSD structure into a series of pseudo sub-systems that are more easily understood and analyzed individually by simplified (non-computerized) methods. This led to the identification of a series of simple mechanistic “systems” that contribute to stresses in the panel. Once the stresses are calculated from analysis on each of these systems, they can be combined by the principle of linear superposition for verification of the applicable design limit states. The same basic approach to describing fundamental behavior is applied herein, with some extensions. Many of the behavioral systems identified by the pioneers in the field are still valid today. In addition, research conducted in recent decades has brought to light many local distortion mechanisms that were not originally apparent. The relevant behavioral mechanisms in the OSD when subjected to wheel loads are summarized in table 4-1 and discussed in detail in the following sections.

Table 4-1 Orthotropic Steel Deck Deformation Mechanisms

System	Action	Figure	Result
1	Local Deck Plate Deformation		Transverse flexural stress in deck and rib plates and RD
2	Panel Deformation		Transverse deck stress from rib differential displacements
3	Rib Longitudinal Flexure		Longitudinal flexure and shear in rib acting as a continuous beam on flexible FB supports
4	Floorbeam In-plane Flexure		Flexure and shear in FB acting as beam spanning between rigid girders
5	Floorbeam Distortion		Out-of-plane flexure of FB web at rib due to rib rotation
6	Rib Distortion		Local flexure of rib wall due to FB cut-out
7	Global		Axial, flexural, and shear stresses from supporting girder deformations

4.2.1. Local Deck Plate Deformation – System 1

Load distribution begins with transfer of local wheel loads from the deck plate to the supporting rib walls. The response is influenced by the spacing of the rib walls and relative thickness (flexural stiffness) of the deck plate and ribs. It is noted that the stresses generated from this mechanism are localized and thus sensitive to the size of the wheel patch load and any load dispersion that may occur through the wearing surface. Very often, it is the front axle of the design truck with only a single tire that can maximize response from this mechanism. It is also noted that this system is one driving factor in the fatigue of the RD, but is generally not a concern for strength based limit states. Simplified analysis of System 1 stresses can be accomplished by employing a finite strip (frame) model of the OSD cross-section in the transverse direction with assumed rigid supports for the ribs, or elastic flexible supports based on the rib flexural stiffness and span length.

4.2.2.Panel Deformation – System 2

By far, the most complicated system to analyze is System 2. Analysis of this system requires an understanding of the two-way load distributing behavior of the OSD panel when subjected to out-of-plane loading, which is a complex problem. Early engineering solutions that used OSDs had an ideological underpinning to the theory of elasticity of plates (plates loaded normally to the plane of the plate). This solution was founded in Huber's Equation:

$$D_x \frac{\partial^4 w}{\partial x^4} + 2H \frac{\partial^4 w}{\partial x^2 \partial y^2} + D_y \frac{\partial^4 w}{\partial y^4} = p(x, y) \qquad (4\text{-}1)$$

This equation represents the static equilibrium of a plate of uniform thickness with orthogonal and torsional properties,

where:

D_x	=	plate flexural rigidity in the x-direction
D_y	=	plate flexural rigidity in the y-direction
H	=	effective torsional rigidity of the plate
p(x,y)	=	the loading at any point on the plate with coordinates of (x, y)

To solve for the stresses, presuming D_x, D_y, and H can be estimated, it is necessary to find w(x,y), the vertical displacement in the z direction that satisfies the homogenous and particular solution of the loading p(x,y). The moments per unit length are then given by:

$$M_x = -D_x \left(\frac{\partial^2 w}{\partial x^2} + \nu_y \frac{\partial^2 w}{\partial y^2} \right); \qquad (4\text{-}2)$$

$$M_y = -D_y \left(\frac{\partial^2 w}{\partial y^2} + \nu_x \frac{\partial^2 w}{\partial x^2} \right) \text{; and} \qquad (4\text{-}3)$$

$$M_{xy} = 2D_{xy} \frac{\partial^2 w}{\partial x \partial y} \qquad (4\text{-}4)$$

where:

D_{xy}	=	$\frac{1}{4}\left(2H - \nu_x D_x - \nu_y D_y\right)$,
ν	=	Poisson ratio in each respective direction.

Stresses are then derived from cross-section properties per unit length. Huber's equation is derived with the assumption that axial forces in the plane of the plate are not present.

The differential equation lends itself to solutions by Fourier Series. Rigorous solutions of this equation for OSDs are truly daunting tasks when FBs represent either discontinuities in the continuum or uncertain boundaries with which compatibility is necessary, but whose displacement is not easily understood. Recently, Higgins (2003, 2004) successfully applied this solution technique for calculation of moments and deflections in filled grid deck systems, which

is supported by FEA and experimental results. However, OSDs with closed ribs have added complications due to increased torsional rigidity of the ribs. Single closed ribs must be smeared into a uniform property of an idealized plate. This is an idealization whose errors are not known. The most widely accepted solution to this equation was first proposed by Pelikan and Esslinger (1957).

The pioneers who developed these analytical techniques were practical individuals who knew how to make reasonable assumptions in an era when little was known about fatigue and the aim was to determine conformance to allowable stresses for service evaluation, in which even large local error is not important, because of the high ultimate strength of the deck. As expressed in the Design Manual for Orthotropic Steel Plate Deck Bridges (AISC, 1963), the simplifications attributed to Esslinger and Pelikan are as follows:

Open Ribs

- The deck plate is treated as a beam, i.e., the plate is given rigidity in the short direction, from rib to rib. Deflection and flexure (at 178.7 MPa [25.9 ksi]) and shear criteria governed, giving a 9 mm (3/8 inch) thickness over a 305 mm (12 inch) rib spacing, for a 53.4 kN (12 kip) wheel load.
- The wheel load is distributed to adjacent ribs as in a beam on elastic foundations.
- Effective width of deck plate (used to calculate the rib/deck plate composite properties over major rib carrying load) is a function of its share of the wheel load and of the "effective" rib span. The effective width is usually larger than the actual rib spacing. The effective rib span is always 0.7 times the actual span.
- Ribs "near" the FB support are treated as resting on rigid foundations, and ribs "near" FB mid-span are treated as resting on flexible foundations. Ribs near mid-span have larger positive moments and smaller negative moments than those near FB support. The Design Manual for Orthotropic Steel Plate Deck Bridges (AISC, 1963) gives moment relief formulae, based on sinusoidal deflection of the FB.
- In short, concepts of orthotropy are abandoned in favor of partial compatibility between beams. Global transverse rigidity is ignored; influence lines for beams are invoked.

Closed Ribs

- The torsional rigidity of the deck plate is governed by G, K, and μ as defined by the following equation:

$$H = \frac{1}{2}\left(\frac{\mu G K}{a+e}\right) \tag{4-5}$$

- Where G is the shear modulus for steel, K is a factor representing the physical properties and geometries of the rib such:

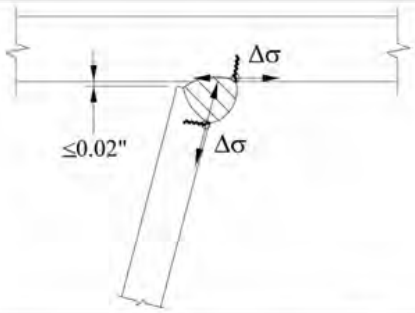
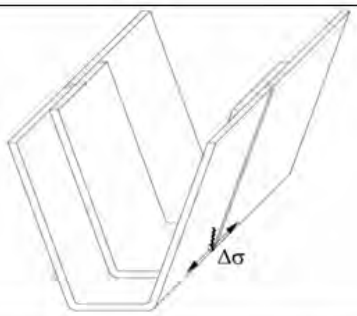
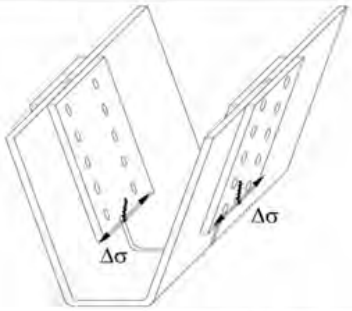
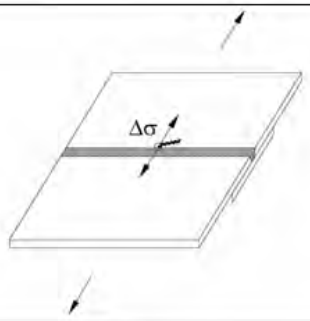
$$K = \frac{4A_r^{\,2}}{(u/t_r)+(a/t_p)} \tag{4-6}$$

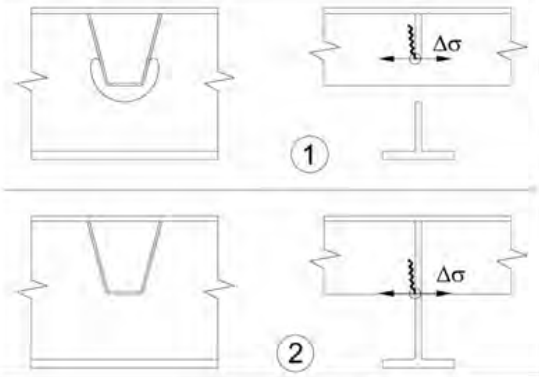
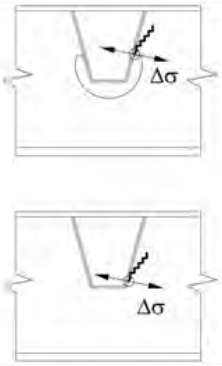
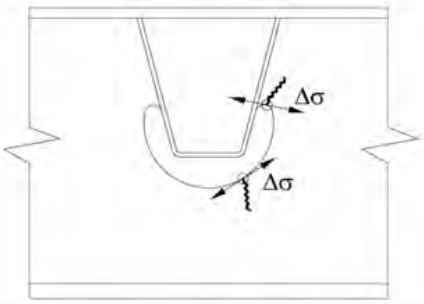
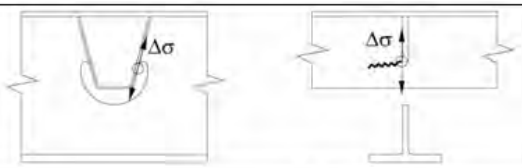
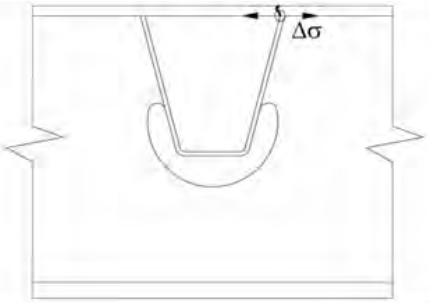
control of RDF root cracking since the notch condition is different from that at the weld toe. Also, little known research has been done to quantify the performance of the detail and parameters of influence. This should be the subject of future research efforts. Until then, Level 3 can be applied, but with additional verification to best extent reasonably possible.

5.7.7. Summary

A summary of the fatigue design requirements for fatigue-prone orthotropic panel details, including illustrative examples, is shown in Table 5-1.

Table 5-1 Summary of Fatigue Design Requirements for OSD Details

Detail	Illustrative Example	Description of Condition	Fatigue Category Detail	Allowable Design Level
Rib-to-deck (RD) Weld		One-sided 80% (70%min) penetration weld with root gap ≤ 0.5mm (0.02") prior to welding	C	1,2, or 3
Rib Splice (Welded)		Single groove butt weld with permanent backing bar left in place. Weld gap > rib wall thickness.	D	1,2, or 3
Rib Splice (Bolted)		Base metal at gross section of high strength slip critical connection.	B	1,2, or 3
Deck Plate Splice		Transverse or Longitudinal single groove butt splice with permanent backing bar left in place.	D (in plane) C (out of plane)	1,2, or 3

Rib to FB (RF) Weld (Rib)		Rib wall at rib to FB weld (fillet, PJP, or CJP).	C	1,2, or 3
Rib to FB (RF) Weld (FB Web)		FB web at rib to FB weld (fillet, PJP, or CJP)	C*	1 or 3
FB Cut-out		Base metal at edge with "smooth" flame cut finish as per AWS D1.5.	A	1 or 3
Rib Wall at Cut-out		Rib wall at rib to FB weld (fillet or CJP).	C	1 or 3
Rib-to-deck Plate at FB (RDF)		Base metal of deck at root of rib-to-deck fillet weld	C	1 or 3

***** Where stresses are dominated by in-plane component at fillet or PJP welds, Figure 5-6 shall be considered. In this case, stress should be calculated at the mid-thickness and extrapolation procedure need not be applied.

are two types of installations, slurry or multiple-layer. The first method involves using slurry with mixed-in fine aggregate, which is applied with a squeegee onto a clean deck and coarse aggregate is applied prior to drying. The second method is constructed in layers of epoxy and cast aggregate. Both surface types contain epoxies with "flexibilizers," or urethanes that render them less brittle.

For further discussion of properties, advantages, and disadvantages of alternate substrates and of epoxy concrete, the reader is referred to Chapter 9.

Bottom of Deck

Protection of the bottom of OSDs is typically achieved with the use of a paint coating system. Painting technology for bridge structures is an ever-evolving science. The FHWA has ongoing studies at the Turner-Fairbank Highway Research Center that will provide key findings, not only on preparation and coating performance, but on corrosivity of environments, environmental compliance, toxicity and containment, etc. Preliminary findings are that three-coat systems in which zinc based primer, epoxy coat, and urethane finish coat work the best. Various state DOTs and bridge owners will derive from these studies their own standards and practices.

Paint systems are most often applied in the shop and touched up in the field where damaged by transportation and handling, or over field-welded areas. OSDs are optimal structures for resisting corrosion, as they have smooth surfaces and no perches, such as bottom flanges for moisture to pond or for birds to nest.

Rib Protection

The rib needs to be protected internally from corrosion due to condensation. The common practice in the U.S. is to place internal steel bulkheads (sealing plates) on each side of the bolted splice to completely seal the ribs from moisture penetration. These are welded all around or welded along the rib and caulked at the top. This prevents moist air from entering the inside of the closed rib. In the area of the splice, the ribs are painted internally.

Interior of OSD Box Girder

Where orthotropic ribs stiffen the walls of a box girder, the interior of the box girder needs to be protected internally from corrosion due to condensation. This can be accomplished with a coating, or by a dehumidification system, which has been done successfully overseas in many cases. Since the entire interior space is protected by this system, the internal sealing diaphragms in the ribs at the splices are not needed and maintenance of the interior surfaces is not required. Inspection of welds is easier because the welds are not painted and the access is provided without disturbance of the traffic below or on top of the bridge.

6.3. GENERAL PROPORTIONS

Orthotropic bridge fabrication is labor intensive, and the design should not strive solely to minimize material. As discussed in previous chapters, this has been one of the leading causes of fatigue and wearing surface failure in early designs. Rather, the design should strive for repetition in details, lending itself to automation and advanced technology, such as automatic

welding processes and robotics to maximize economy. OSDs require significantly more shop welding than any other steel bridge system. Strictly speaking, there is nothing difficult about fabrication of OSD panels, compared to other steel bridge construction, except for a singular welding issue for the RD connection and time consuming cut-out details, if they are required. Erection of OSD panels in deck replacements is now commonplace, even if strict field welding rules must be applied.

The recommended limits for panel detailing proportions are summarized in Table 6-1.

In box girders designed for suspension bridges, where low resistance to cross winds is necessary and sharp corners are part of the outer edges of the boxes, close attention must be paid to access for welding of stiffeners and shell plate corners.

Table 6-1 Recommended Limits for Orthotropic Panel Proportions

Detailing Dimension	Limit
Deck Plate Thickness	$t_d > 14$ mm (5/8 inch)
Rib Thickness	6 mm $< t_r <$ 12 mm (1/4 $< t_r <$ 1/2 inch)
Rib Spacing – direct wheel load	600 mm $< s <$ 762 mm (24 $< s <$ 30 inches)
Rib Spacing – no direct wheel load	600 mm $< s <$ 1000 mm (24 $< s <$ 40 inches)
Floorbeam Spacing	$L <$ 6000 mm (20 ft)
Ratio of Rib-to-Floorbeam Depth	$h_{rib}/h_{FB} < 0.4$
Floorbeam Web thickness	10 mm $< t_{FB} <$ 20 mm (3/8 $< t_{FB} <$ 3/4 inch)
Ratio of Cut-out to Rib Depth	$h_{cutout}/h_{rib} > 0.33$

6.4. RIB-TO-DECK PLATE (RD) WELD

Historically, the RD weld has been specified as a one-sided, partial penetration weld with 80 percent minimum penetration and limited melt-through. Because of inherent variations in components when ribs are thin, weld size has tolerances that are of a high magnitude relative to rib thickness. Achieving a minimum of 80 percent penetration without melt-through is difficult. Fabricators, fearful of melt-through, have chosen electrode angles of 60 degrees with the deck plate, without preparing a bevel in the rib, and have often failed to achieve 80 percent penetration. The primary factors that are known to influence the fatigue resistance of the joint are: penetration, melt-through, and root gap.

Penetration

It is not known exactly where the “80 percent rule” came from and hence, it appears somewhat arbitrary. A review of the literature suggests that it was more of a practical limit, i.e., the maximum penetration that could be achieved without regularly resulting in weld melt-through. Kolstein’s review of existing test data suggested that the weld throat should be, at a minimum, of the same size as the rib wall and that the amount of penetration should be between 50 to 80 percent (Kolstein, 2007). This statement must be used with caution, however, as the influence of the lack of fusion zone size is not constant for different plate thicknesses with constant weld

reinforcement size. In other words, 50 percent penetration in a 9 mm (3/8 inch) thick rib wall would result in a different fatigue resistance than a 50 percent penetration weld in a much thicker rib wall with the same weld reinforcement. However, for typical values of rib wall thickness used in highway bridges, the range would seem reasonable.

Although Kolstein suggests a lower limit of 50 percent, this can result in a rather large lack of fusion plane and greater penetration is recommended. Accordingly, penetration levels between 70 to 95 percent, with a target of 80 percent, seem more reasonable. The lower bound of 75 percent is from FEA analysis by Xiao, which indicated that increasing the penetration above 75 percent did not result in any significant improvement in fatigue resistance of the joint (Xiao et al 2008), though no experimental work was performed as part of the study. Nevertheless, all of the data suggest that a range for the degree of penetration is appropriate and that requiring 80 percent is not justified.

Melt-through

There are legitimate concerns that weld melt-through could create undesirable discontinuities at the root that could become the source of fatigue crack initiation, and therefore it should be avoided (Ya, 2011). Fatigue tests conducted by Sim and Uang (2007) at the University of California at San Diego, on behalf of Caltrans, however, indicate that melt-through is not a real problem, even when intentionally introduced after 80 percent penetration, to create abrupt transitions. This suggests that, using SAW, weld size limits between an 80 percent minimum penetration and a maximum amount of melt-through should be a workable and acceptable fabrication procedure, even for 8 mm (5/16 inch) rib thickness.

Root gap

Recent research has shown that fatigue resistance of the weld is clearly improved when the root gap is closed in the final condition (Wright, 2010). Shop experience indicates that using a tight fit prior to welding will also help prevent weld melt-through with a limit of 0.5 mm (0.020 inch).

As a result, the consensus is that the proposed detailing criteria for the weld between the webs of a closed rib and the deck plate be a one-sided, nominal 80 percent penetration, with 70 percent minimum and no blow-through, and with a tight fit less than 0.5 mm (0.020 inch) prior to welding. Additional fabrication techniques and tolerances are discussed in Chapter 7.

6.5. RIB-TO-FLOORBEAM (RF) CONNECTION

At the rib-to-floorbeam (RF) connection, fatigue cracks have been observed in the rib wall and FB at weld toes and weld terminations at cut-outs. The detailing must be made to prevent such crack growth. This detail is the most labor-intensive of all the details, and is where the designer is free to choose the geometry. Cracks at the RF are generally due to the presence of high out-of-plane stress ranges due to distortion of the rib wall. The distortion comes from the FB pumping the rib wall in and out-of-plane. Where cut-outs are used, the problem can be likened to web gap cracking at transverse stiffeners in a plate girder. In order to prevent such cracking, internal FBs or bulkheads have been used to ensure the geometry of the rib is maintained, and hence the distortion minimized or eliminated. However, bulkheads may create other problems (see Section 4.1 for more discussion).

the width of the tooth at the deck. The lower prismatic member has a width equal to the width at mid-depth of the tooth. The effective width of deck plate and longitudinal rib used in the Vierendeel model is illustrated in Figure 11-11.

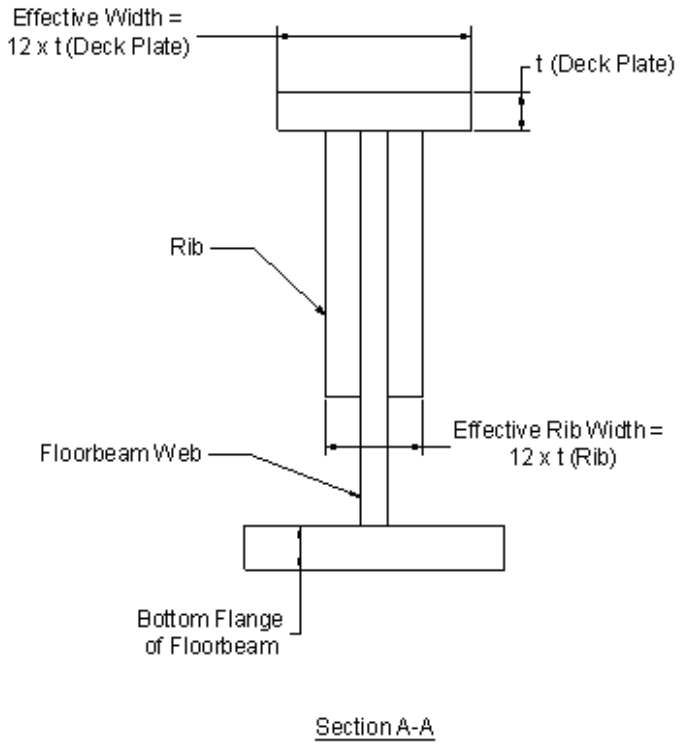


Figure 11-11 Illustration of Effective Width of Deck Plate and Rib Used in the Vierendeel Model

To verify the FEA results, a 445 kN (100 kips) vertical point load is applied directly above the eighth tooth in the FB (which corresponds to 0.33L of span 2, where L = span length). This is shown schematically in Figure 11-12 and Figure 11-13. First, horizontal moments acting on two of the most highly-loaded teeth are checked. The moment on a given tooth is checked at a section of the tooth right before the rib begins to "radius" (as illustrated in Figure 11-12). The FEA moment in a tooth is determined by creating a horizontal slice through a tooth and integrating the stresses over the tooth sectional area to arrive at the net moment on the section. A comparison of the Vierendeel and FEA teeth moment is shown in Table 11-1. As can be seen from the table, the ratios of Vierendeel-to-FEA moments are 0.90 and 0.94 in Tooth 7 and Tooth 10, respectively.

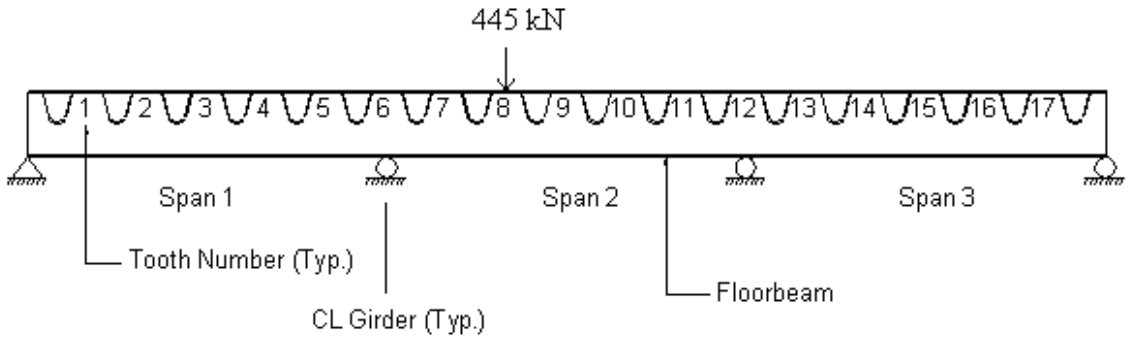


Figure 11-12 Floorbeam Elevation Showing the Verification Load, Tooth Numbering, and the Girders Represented as Supports for the Model

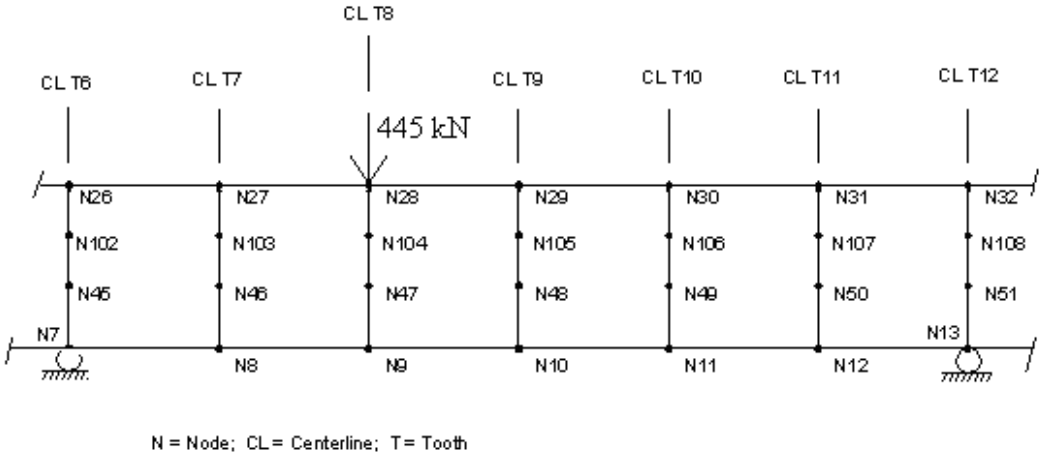


Figure 11-13 Span 2 of the Vierendeel Model showing Verification Load, Node Numbers, and Tooth Numbers for the Model

Table 11-1 Comparison of Vierendeel and FEA Tooth Moments

Vierendeel Node No.	Floorbeam Tooth No.	A = Vierendeel Moment (kN-m)	B = FEA moment (kN-m)	A ÷ B	Location Along Floorbeam (L = Span Length)
N46	7	-48.0	-53.3	0.90	0.17L of Span 2
N49	10	28.2	30.1	0.94	0.67L of Span 2

Next, vertical stresses acting on a horizontal section through the teeth is checked at the same section where the horizontal teeth moments were checked (shown in Figure 11-14). A comparison of the Vierendeel and FEA tooth stress (in two teeth) is shown in Table 11-2. On the left side of Tooth 7, the Vierendeel stress is 18 percent higher compared to the FEA stress. On the right side of Tooth 7, and on the left and right sides of Tooth 10, the Vierendeel and FEA stresses are within 5 percent of each other. A schematic of the FEA vertical stress contours in the FB is shown in Figure 11-15. It shows that the maximum tensile and compressive vertical

stresses, excluding the stress directly below the applied concentrated load, occur at the left and right edges of the teeth, near the base of the teeth where the ribs begin to “radius”.

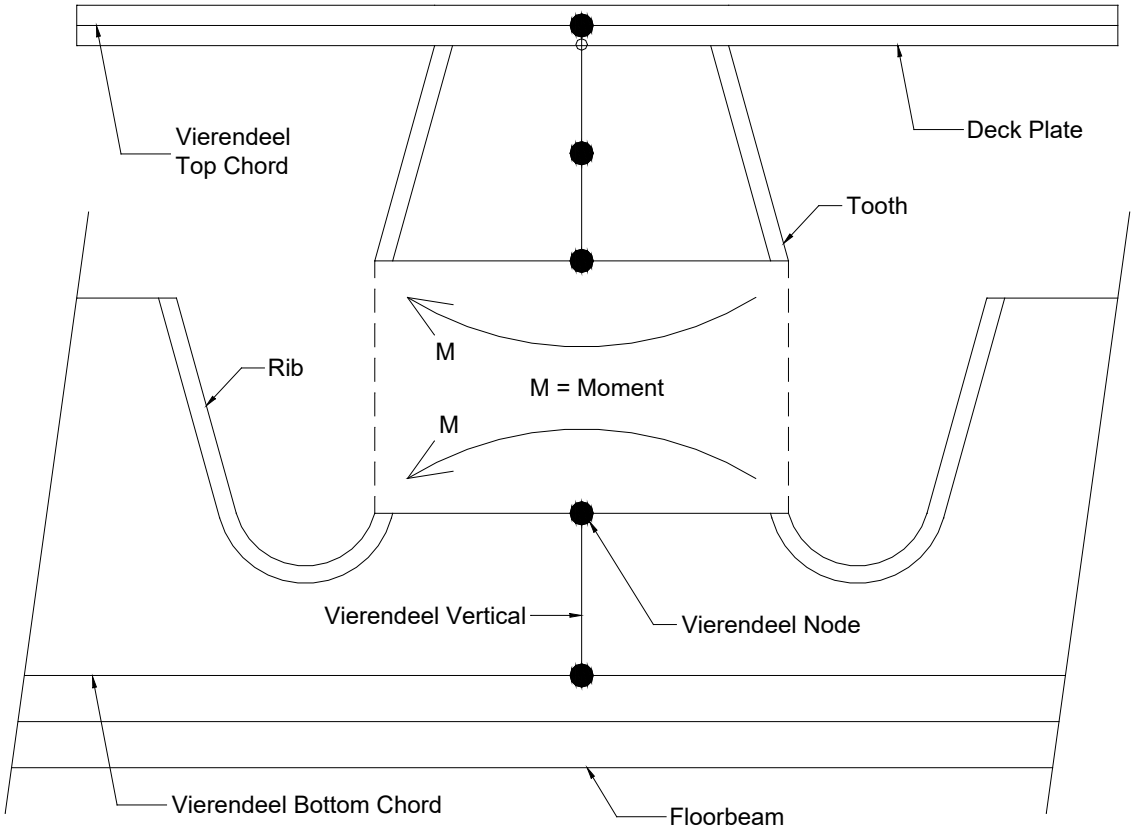


Figure 11-14 Illustration of Horizontal Moment Acting at the Base of a Tooth and at a Vierendeel Node Superimposed on the Floorbeam

Table 11-2 Comparison of Vierendeel and FEA Tooth Stresses

Vierendeel Node No.	Floorbeam Tooth No.	A = Vierendeel Min. Stress (MPa)	B = FEA Min. Stress (MPa)	A ÷ B	C = Vierendeel Max. Stress (MPa)	D = FEA Max. Stress (MPa)	C ÷ D
N46 (bottom)	7	-86.81	-73.64	1.18	86.81	91.63	0.95
N49 (bottom)	10	-50.95	-53.09	0.96	50.95	50.06	1.02
Negative Stress (-) is Compressive.							

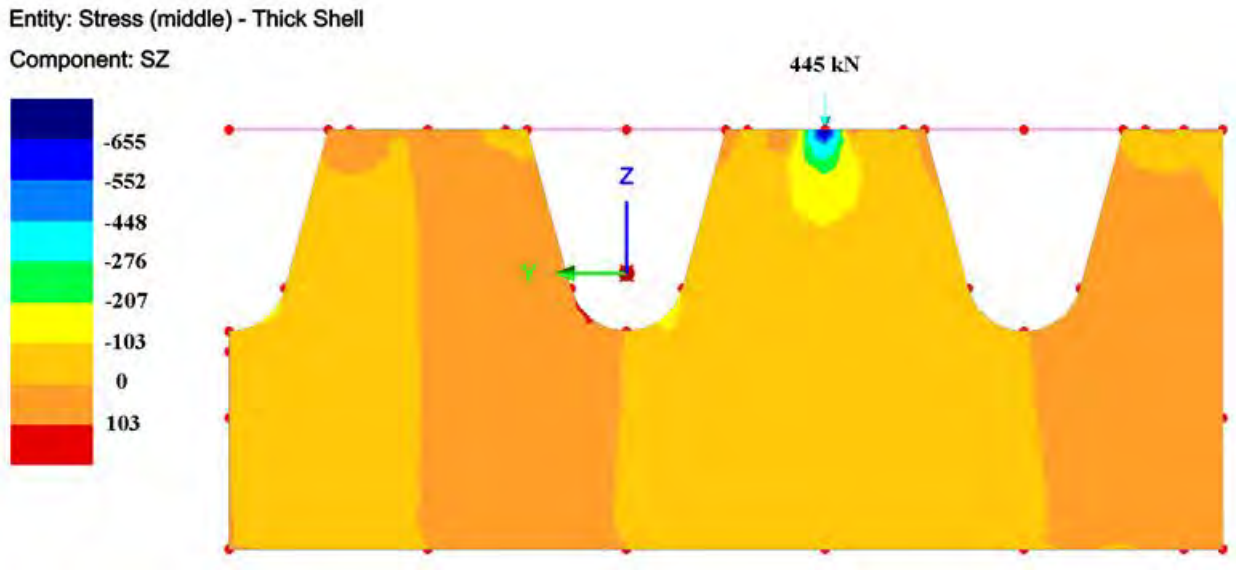


Figure 11-15 Results of FEA Model of the Floorbeam showing the Vertical Stress Contours in the Web and Tooth

The absolute maximum FEA tensile stress in the tooth is 99.63 MPa, (22.4 kips) and occurs on the right side of Tooth 7 along the curved portion at the base of the tooth. This stress is 15 percent higher than the absolute maximum Vierendeel tensile stress in the teeth (86.81 MPa [19.5 kips]). The absolute maximum compressive stress in the tooth is 81.50 MPa (18.3 kips), and occurs on the left side of Tooth 7 along the curved portion at the base of the tooth. This stress is 6 percent lower than the absolute maximum Vierendeel compressive stress in the teeth (86.81 MPa [19.5 kips]).

11.1.3.2. Bending in Rib at the Floorbeam

In the preliminary design of this example, the FB spacing was set at 4,572 mm (15 ft) before being revised to 3,048 mm (10 ft). This section contains a check that was performed on a model with the original 4,572 mm (15 ft) FB spacing, but the conclusions reached are just as valid for the final design.

An estimate of the bending moment in a rib at a FB can be made by assuming a distribution factor and a rigid support at the FB. A distribution factor of 0.5 for closed ribs has been shown to often result in conservative moments.

A single wheel load of 71.2 kN (16 kips) is distributed to a single rib using a distribution factor of 0.5, resulting in an applied load of 35.6 kN (8 kips). This load is spread along a patch of 250 mm (10 inches) in length. The equations and methods outlined in Design Manual for Orthotropic Steel Plate Deck Bridges (AISC, 1963) are used to determine the resulting moment in a rib. The equation given below is used to calculate the moment in a rib at a support when a distributed load is present 0.3804 times the FB spacing away.

$$\frac{M}{SP} = -0.085 + 0.1494\left(\frac{C}{S}\right)^2 \qquad (11\text{-}1)$$

where:

M	=	moment in rib
S	=	Spacing of FBs
C	=	½ Patch load length
P	=	Total applied force

For the present case, S = 15’ (4572 mm), C = 5” (127 mm) = 0.4165’ (126.95 mm)

$$\frac{C}{S} = \frac{0.4165'}{15'} = 0.0278$$

$$\frac{M}{SP} = -0.085 + 0.1494(0.0278)^2 = -0.0849$$

$$M = 15' \times 8\ kips \times -0.0849 = -10.2\ kip \cdot ft = -13.8\ kN \cdot m$$

The same loading is applied in the Finite Element model, with the patch load centered on one of the two center ribs in a panel. The loading is applied at 0.3804 times the panel length away from the FB. The moment in the rib is found by integrating the stresses in the shell elements comprising the rib and a portion of the deck extending half-way to the adjacent rib. The integrated moment is found to be -14.8 kN-m (-10.9 kip-ft). This is in very good agreement with the value calculated by the method given in the AISC publication.

11.1.4. Service Limit States Checks

As specified in Section 5.2.2 of this manual, the AASHTO Service I limit state must be satisfied for overall deflection limits of the deck plate, the ribs, and relative deflection between adjacent ribs. The relative deflection limit is set as 2 mm (0.1 in.) and is checked at the mid-span of the ribs between the shell modeled FBs. Dead load and live load are applied to the model, with the live load being an unfactored HL-93 design truck placed at various transverse locations with the rear axle centered at the mid-span between the FBs.

Enveloping the load cases considered, the maximum relative deflection is found to be roughly 0.5 mm (0.02”), which is well below the allowable limit.

11.1.5. Strength Limit State Checks

The negative moment capacity of a rib located between the girders at an interior support is checked. The loading used to create the maximum negative moment at the RF consists of an AASHTO design truck with a 4300 mm (14 ft) rear axle spacing and the 9.34 kN/m (0.64 klf) notional live loading applied across a lane width of 3048 mm (10 ft). The truck is placed so that the rear set of axles are equidistant from the FB. This loading can be placed at various transverse locations to produce maximum moments at different ribs. As specified in this manual, the 142

kN (32 kip) axles are split into two 71 kN (16 kip) axles spaced 1220 mm (4 ft) apart. A pressure of 275 kPa (40 psi) is used for the two front tires, which are each a 250 mm (10 inch) square patch, and also for the remaining eight rear tires, which are each a 510 mm (20 inch) wide x 250 mm (10 inch) long patch.

The effective width of deck plate to use is found in Section 4.3of this Manual. An OSD with closed ribs is classified as a non-composite box section in AASHTO LRFD, and Eq. 6.12.2.2.2-1 gives the equation for M_n. Since the ribs can not buckle laterally, the equation results in M_n being equal to the moment at first yield.

Examining the longitudinal stresses at the bottom of the rib (the extreme fiber in bending) under appropriate load factors, it is apparent that all of the stresses, even the concentrations due to the connection of the rib to the FB, are well below the yield stress. Alternatively, the moments can be extracted from the ribs by integrating over the effective width of the rib, and these moments can be compared with the yielding moment for the section.

11.1.6. Fatigue Limit State Checks

The finite element model developed for Example 1 is analyzed to find the live load fatigue stress range for details of OSDs with ribs without cut-outs. The first five types of details illustrated in Table 5-1 of Chapter 5 are examined and compared with their constant amplitude threshold allowable stress ranges for the applicable category, which can also be found in Table 5-1. An HL-93 design truck with 9000 mm (30 ft) between centerlines of the rear axles groups is used in accordance with the AASHTO LRFD fatigue specifications and the guidelines found in the previous chapters. AASHTO LRFD specifies that the 142 kN (32 kip) axles be split into two 71 kN (16 kip) axles 1220 mm (4 ft) apart as illustrated in Figure 5-1. This results in a uniform tire pressure of 275 kPa (40 psi), which is distributed over a 250 mm (10 inch) square patch for the two 17.75 kN (4 kip) wheels, and over a 510 mm (20 inch) wide x 250 mm (10 inch) long patch for the eight 35.5 kN (8 kip) wheels. This loading model is altered slightly to account for the distribution of the load through the wearing surface and one half of the deck plate, as allowed in Section 5.5.1.1 of this manual. At the ends of each tire edge, an amount equal to the distance from the top of the wearing surface to the mid-depth of the deck is added, and the pressure is then altered to result in the correct total load for the wheel. An impact factor of 15 percent is applied to all loadings, as well as the infinite life fatigue load factor of 2.25 where applicable (see Section 5.3).

Table 11-3 summarizes the results from the analysis, which will be described in the subsequent sections.

Table 11-3 Summary of Fatigue Checks for Example 1

Detail	Point of interest	S_r (MPa)	$S_{r,allow}$ (MPa)
RD (Deck)	Deck at rib connection directly under load mid-span between FBs.	55.9	69.0
RD (Rib)	Rib at deck connection opposite of loaded side mid-span between FBs.	33.9	69.0
Rib Spice (Max)	Lowest point of rib at 0.3 of the span between FBs.	34.3	48.0
Rib Spice (Min)	Lowest point of rib at 0.3 of the span between FBs.		
Deck Plate Splice	Deck splice midway between ribs at mid-span between FBs.	40.0	69.0
RF (Rib)	Lower half of rib at FB connection	39.5	69.0
RF (FB.)	FB at rib-to-deck connection and at base of rib	28.3	69.0

11.1.6.1. Rib-to-Deck (RD) Weld

To get the maximum stress range in the rib-to-deck weld (RD), a single static loading of one rear wheel patch is applied. The wheel load is centered on one leg of a rib, as illustrated in Figure 11-16, and applied at mid-span of the panel between FBs (note the width of the wheel patch is shown, not the distribution of the load through the wearing surface and one half of the deck plate). A schematic showing the longitudinal position of the loading is shown in Figure 11-17. The locations of interest are the deck plate at the rib connection directly under the load, and the rib at the deck connection in the rib leg opposite the load (see Figure 11-18 and Figure 11-19).

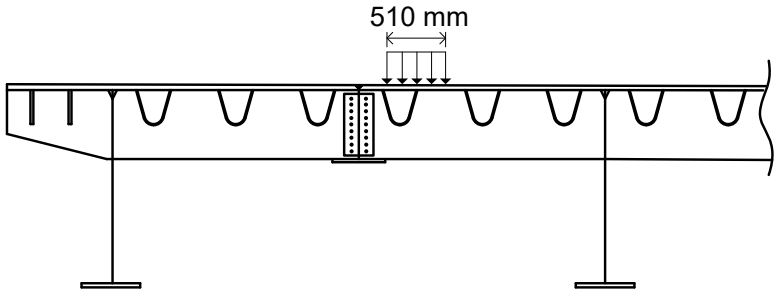


Figure 11-16 Transverse Wheel Location to Produce the Largest Impact on the Rib-to-deck Weld

$$F_{cr} = 0.91\left[0.658^{\frac{0.91\times 345\,MPa}{2100\,MPa}}\right]345\,MPa = 295\ MPa$$

$$\phi P_n = \phi(A \times F_{cr}) = 0.9(195.6\,cm^2 \times 295\,MPa \times \frac{1\,kN}{1000\,N} \times \frac{(10\,mm)^2}{(1\ cm)^2}) = 5190\ kN$$

The dead and live load axial forces in the deck plate (per rib strut) were given to be 1,025 kN (230 kips) and 380 kN (85 kips), respectively. Using the Strength I load factors, the factored load is 2,000 kN (440 kips) for each strut, which is well below the calculated φP_n value of 5190 kN (1170 kips).

11.2.5. Fatigue Limit State Checks

The finite element model developed for Example 2 is used to find the live load fatigue stress ranges for details in OSDs with relief cut-outs at four locations. The allowable stress ranges for the various details are calculated based on finite life of the details, using the given (ADTT)SL, a 75-year design life, and n (cycles per truck passing) equal to 1. Note that this example is calculating the fatigue life for details not directly connected to the deck plate. Those details connecting to the deck plate experience one cycle per axle, i.e. n = 5. See Chapter 5 for additional information on the number of cycles.

As in Example 1, a design truck with a constant 9,000 mm (30 ft) between rear axles groups is used in accordance with the AASHTO LRFD fatigue specifications and the guidelines in the previous chapters. The 142 kN (32 kip) axles are split into two 71 kN (16 kip) axles 1,220 mm (4 ft) apart, as illustrated in Figure 5-1. This results in a uniform tire pressure of 275 kPa (40 psi), which is distributed over a 250 mm (10 inch) square patch for the two 17.75 kN (4 kip) wheels, and over a 510 mm (20 inch) wide x 250 mm (10 inch) long patch for the eight 35.5 kN (8 kip) wheels. This loading model is altered slightly to account for the distribution of the load through the wearing surface and one half of the deck plate, as allowed in Section 5.5.1.1 of this Manual. At the ends of each tire edge, an amount equal to the distance from the top of the wearing surface to the mid-depth of the deck is added, and the pressure is then altered to result in the correct total load for the wheel.

The truck is moved along the model in 610 mm (2 ft) increments, with the wheel centered over an interior rib to generate an influence line for the stresses. After determining the approximate longitudinal positions of the wheels that resulted in the highest stresses, the truck is shifted transversely so the wheels are centered over the rib, the rib wall, and the tooth of both an interior rib and the edge rib for a limited number of longitudinal positions. From these loadings the position of the truck that resulted in the largest stress for each of the four details is determined. An impact factor of 15 percent is applied to all loadings, as well as the finite life fatigue load factor of 0.75.

Table 11-4 summarizes the results of the analysis, with explanations of the calculations given in the subsequent sections.

Table 11-4 Summary of Fatigue Checks for Example 2

Detail	Point of Interest	S_r (MPa)	$S_{r,allow}$ for $(ADTT)_{SL}$ = 485 (MPa)
RF (Rib)	At start of cut-out	38.6	47.7
Rib Wall @ Cut-out	At cut-out	47.7	47.7
RF (Floorbeam)	At start of cut-out	17.9	47.7
Floorbeam Cut-out	Around free edge of cut-out	65.3	85.2

The calculation of the allowable stress ranges for the two types of details is performed using Section 6.6.1.2.5 of the AASHTO LRFD Bridge Design Specifications (AASHTO 2010). The calculations are shown below.

For the Category C details (RF, rib wall): AASHTO Ref.

$A = 44 \times 10^8 \ ksi$ Table 6.6.1.2.5-1

$N = 365(75)n(ADTT)_{SL} = 365(75)(1)(485) = 13{,}280{,}000$ Equation 6.6.1.2.5-3

$$\Delta F = \left(\frac{A}{N}\right)^{1/3} = \left(\frac{44 \times 10^8}{13{,}280{,}000}\right)^{\frac{1}{3}} = 6.92 \ ksi = 47.7 \ MPa$$ Equation 6.6.1.2.5-2

For the Category A detail (FB cut-out):

$A = 250 \times 10^8 \ ksi$ Table 6.6.1.2.5-1

$N = 13{,}280{,}000$

$$\Delta F = \left(\frac{A}{N}\right)^{1/3} = \left(\frac{250 \times 10^8}{13{,}280{,}000}\right)^{\frac{1}{3}} = 12.35 \ ksi = 85.2 \ MPa$$ Equation 6.6.1.2.5-2

11.2.5.1. Rib-to-Floorbeam (RF) Weld – Rib Detail

The maximum longitudinal stress in the rib at the FB web occurs in the edge rib with the truck positioned so the wheel load is located on the tooth (as seen in Figure 11-42a), with the FB positioned halfway between the front axle and the first middle axle, as illustrated in Figure 11-43. The maximum stress range occurs in the rib at the termination of the FB web cut-out, as illustrated in the stress contour of Figure 11-38. The stresses are in the element local y-direction, which is the longitudinal direction of the rib.

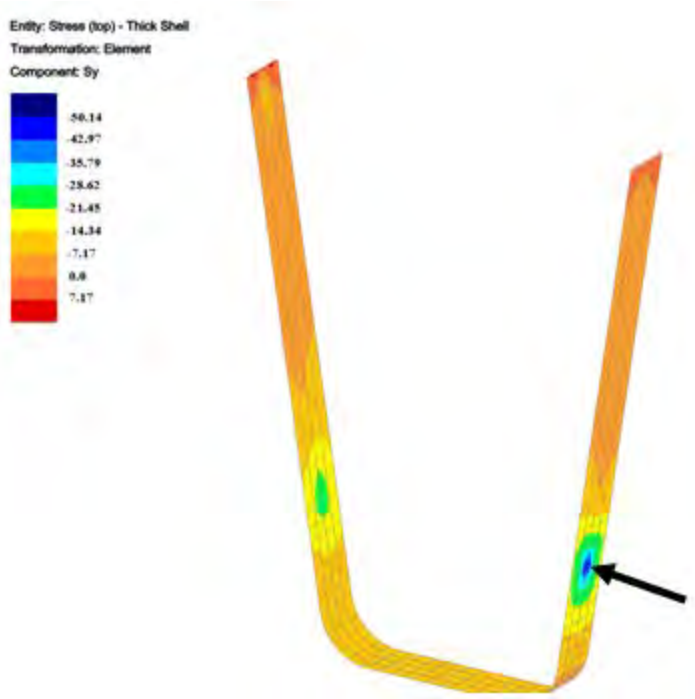


Figure 11-38 Results of FEA Model of the Rib to Floorbeam Weld showing the Rib portion of the Model for the Longitudinal Stress Contours in Rib Wall (in MPa)

Due to the stress concentration seen in Figure 11-38, the stress extrapolation technique is used to find the unfactored force effect. This value is then multiplied by the appropriate factors to determine the factors' live load stress range, as seen below.

$$\sigma_r = 1.5(-39.6\ MPa) - 0.5(-29.3\ MPa) = -44.75\ MPa$$

$$\gamma(\Delta f)_{RIB-LONGITUDINAL} = 0.75(1.15 \times 44.75\ MPa) = 38.6\ MPa$$

Table 11-4 shows that the calculated stress range of 38.6 MPa for this detail is less than the finite life allowable stress range of 47.7 MPa for the anticipated ADTT.

11.2.5.2. Rib Wall at Cut-out

The maximum vertical stress in the rib at the FB web cut-out also occurs in the edge rib with the truck positioned such that the wheel load is located on the tooth, with the FB positioned halfway between the front axle and the first middle axle. Figure 11-39 shows the stress contours and illustrates the location of the stress. The stresses are in the element local x-direction, which is the direction along the perimeter of the rib cross-section.